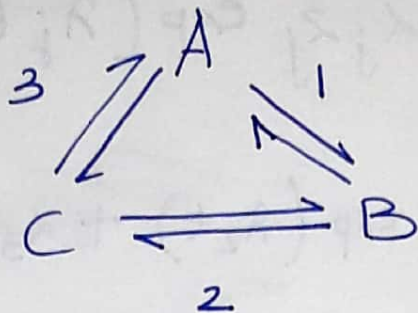




$$\eta_1 = k_{1f}X_A - k_{1b}X_B$$

$$\eta_2 = k_{2f}X_B - k_{2b}X_C$$

$$\eta_3 = k_{3f}X_C - k_{3b}X_A$$

$$\frac{dX_A}{dt} = -\eta_1 + \eta_3 = -k_{1f}X_A + k_{1b}X_B + k_{3f}X_C - k_{3b}X_A$$

$$\frac{dX_B}{dt} = -\eta_2 + \eta_1 = -k_{2f}X_B + k_{2b}X_C + k_{1f}X_A - k_{1b}X_B$$

$$\frac{dX_C}{dt} = -\eta_3 + \eta_2 = -k_{3f}X_C + k_{3b}X_A + k_{2f}X_B - k_{2b}X_C$$

$$\frac{d}{dt} \begin{bmatrix} X_A \\ X_B \\ X_C \end{bmatrix} = \begin{bmatrix} -k_{1f} - k_{3b} & k_{1b} & k_{3f} \\ k_{1f} & -k_{2f} - k_{1b} & k_{2b} \\ k_{3b} & k_{2f} & -k_{3f} - k_{2b} \end{bmatrix} \begin{bmatrix} X_A \\ X_B \\ X_C \end{bmatrix}$$

$$X = X^* + \sum_{j=1}^{n-1} d_j Z_j \exp(\lambda_j t)$$

$$\underline{X} = X^* + d_2 \underline{Z}_2 \exp(\lambda_2 t) + d_3 \underline{Z}_3 \exp(\lambda_3 t)$$

For the chosen time, II term is negligible

Experiment 4
Data

$$\underline{X} = \underline{X}^* + d_3 \underline{Z}_3 \exp(\lambda_3 t)$$

$$X_1 = X_1^* + d_3 Z_{31} \exp(\lambda_3 t)$$

$$X_1 - X_1^* = d_3 Z_{31} \exp(\lambda_3 t)$$

$$\ln(X_1 - X_1^*) = \ln(d_3 Z_{31}) + \lambda_3 t$$

Plot t Vs $\ln[X_1 - X_1^*]$

$$\lambda_3 = -5.625$$

$$d_3 Z_{31} = -3.5157$$

$$\ln(X_3 - X_3^*) = \ln(d_3 Z_{33}) + \lambda_3 t$$

$$\lambda_3 = -7.2357$$

$$d_3 Z_{33} = 0.0494$$

$$\lambda_3 = -6.4303$$

$$Z_{32} = 0$$

$$\underline{Z}_3 = \begin{bmatrix} 1.3505 \\ 0 \\ 2.2977 \end{bmatrix}$$

Take average

$$\cancel{X_2} = \cancel{X_2}$$

Experiment 2 data

$$\text{Average } \lambda_2 = -3.5016$$

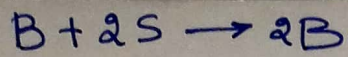
$$\underline{Z}_2 = \begin{bmatrix} -6.9 \\ 0 \\ -4.48 \end{bmatrix}$$

λ_1 & $\underline{Z}_1$
provided
in question

$$\underline{k} = \underline{Z}^{-1} \underline{\Lambda} \underline{Z}$$

$$\underline{Z} = \begin{bmatrix} \underline{Z}_1 & \underline{Z}_2 & \underline{Z}_3 \end{bmatrix} \quad \underline{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

$$\underline{k} = \begin{bmatrix} -3.7233 & -12.3984 & 6.3589 \\ -12.3984 & -112.8960 & 57.9020 \\ 6.3589 & 57.9020 & -29.6768 \end{bmatrix}$$



Given Data : t, C_B, C_S ; $\eta = \frac{\mu C_S C_B}{K_S + C_S}$

Find μ, K_S

$$C_{S0} = 10 \text{ mol/m}^3$$

$$C_{B0} = 0.1 \text{ mol/m}^3$$

Find t, ε_s

$$\frac{d\varepsilon_s}{dt} = \eta$$

$$\frac{d\varepsilon_s}{dt} = \frac{\mu (C_{S0} - 2\varepsilon_s) (C_{B0} + \varepsilon_s)}{K_S + (C_{S0} - 2\varepsilon_s)}$$

$$\frac{d\mu}{dt} = 0$$

$$\frac{dK_S}{dt} = 0$$

$$\frac{d}{dt} \begin{bmatrix} \varepsilon_s \\ \mu \\ K_S \end{bmatrix} = \begin{bmatrix} \eta \\ 0 \\ 0 \end{bmatrix}$$

$$\frac{d\underline{x}}{dt} = \underline{f}$$

Start with the guess

$$\underline{x}^1(0) = \begin{bmatrix} 0 \\ 1e-3 \\ 1e-3 \end{bmatrix}$$

$$\underline{x}^2(t) = \underline{p}(t) + \sum_{j=1}^3 \underline{h}^j(t) x_j^1(0)$$

First component

$$\underline{x}_1^2(t) = \underline{p}_1(t) + h_1^1(t) x_1^1(0) + h_1^2(t) x_2^1(0) + h_1^3(t) x_3^1(0)$$

$$\underline{p}_1(t) = \exp(-\underline{J}_1^* t)^* \left(\underline{f}_1 - \underline{J}_1^* x_1^1(0) - \underline{J}_2^* x_2^1(0) - \underline{J}_3^* x_3^1(0) \right)^* t$$

$$\underline{J} = \begin{bmatrix} J_1 & J_2 & J_3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{d \underline{h}^j}{dt} = \underline{J} \underline{h}^j \bigg|_{x^1(0)}$$

$$\underline{h}^j = e^{t \underline{J}} \underline{h}^j(0)$$

$$h_1^1 = 1 + t J_1$$

$$h_1^2 = t J_2$$

$$h_1^3 = t J_3$$

3
-3

$$\min_{t=1}^n \left\{ \left[\varepsilon_s^{\text{data}}(t) - \varepsilon_s^2(t) \right]^2 \right\} \rightarrow f_{\text{minunc}}$$

$\underline{x}(0)$

↓
find $\underline{x}^2(0)$ at the last t

Use this $\underline{x}^2(0)$ & iterate

~~Convergence~~ Calculate $\varepsilon_s^{\text{sol}}$ by solving

$$\frac{d\varepsilon_s}{dt}$$

$$\frac{d\underline{x}}{dt} = \underline{f} \text{ with } \underline{x}^2(0) \text{ as initial condition}$$

Convergence criteria

$$\left| \varepsilon_s^{\text{data}} - \varepsilon_s^{\text{sol}} \right| < 1e-6$$

Solution : ~~$\mu = 0.08$~~

$$\mu = 0.0612$$

$$K_s = 0.001$$

Start with initial guess

$$\begin{bmatrix} 0 \\ 1e-3 \\ 1e-3 \end{bmatrix}$$

→ Actual solution (0.08, 0.001)

Objective function has local minima.